

MDS220 — Musculoskeletal, CNT & Integumentary System

Full-length Practice Exam · 202 Questions · Mixed Difficulty · Lectures 1–23

Single best answer · Examiner-style · Detailed key at the end

About this paper. Built to mirror your professors' past-paper style — short single-best-answer stems, frequent 'which is NOT / correct pairing / spares' items, and a mix of straight recall and applied/clinical reasoning. Section sizes follow the real exam weighting (Lec 1–3 largest, then 4–6, etc.). Everything is grounded strictly in the uploaded slides. Correct answers are length-balanced and positions shuffled, so you can't guess from format. Note: the real leg-region exam may show an X-ray photo and ask you to identify it — that image format isn't reproduced here.

Section 1 · Anatomical Terms, Connective & Muscular Tissue (Lec 1–3)

1. Which structure is NOT part of the neuromuscular junction?
 - A) Motor end-plate
 - B) Junctional fold
 - C) Synaptic cleft
 - D) Axon terminal
 - E) Intrafusal fibre
2. H&E staining works mainly on the principle of:
 - A) Metal impregnation
 - B) Cationic–anionic affinity
 - C) Antibody binding
 - D) Carbohydrate oxidation
 - E) Enzyme activity
3. Passing fixed tissue through graded ethanol is the step termed:
 - A) Fixation
 - B) Infiltration
 - C) Clearing
 - D) Embedding
 - E) Dehydration
4. With toluidine blue, which cell component is MOST basophilic?
 - A) Euchromatin
 - B) Golgi apparatus
 - C) Mitochondria
 - D) Nucleolus
 - E) Nucleoplasm
5. Which connective-tissue cell shows an eccentric 'clock-face' nucleus?
 - A) Macrophage
 - B) Plasma cell
 - C) Lymphocyte
 - D) Mast cell
 - E) Eosinophil
6. Early-wound fibres that are silver- and PAS-positive are:
 - A) Collagen III
 - B) Collagen IV
 - C) Collagen II
 - D) Collagen I
 - E) Elastic fibres
7. Which collagen type builds the basal lamina?
 - A) Type IV
 - B) Type III
 - C) Type V

- D) Type II
- E) Type I

8. In elastic arteries, elastin of the tunica media is secreted by:

- A) Fibrocytes
- B) Mesenchymal cells
- C) Fibroblasts
- D) Smooth-muscle cells
- E) Endothelial cells

9. Which polyanion lets ground substance form a hydrated gel?

- A) Hyaluronic acid
- B) Chondroitin sulphate
- C) Cholesterol
- D) Ceramide
- E) Keratan sulphate

10. The classic model organ for reticular tissue is the:

- A) Spleen
- B) Lung
- C) Thymus
- D) Kidney
- E) Liver

11. Articular cartilage and bursae are typical of which joint type?

- A) Synovial (knee) joint
- B) Suture
- C) Gomphosis
- D) Pubic symphysis
- E) Syndesmosis

12. Which is NOT a function of synovial fluid?

- A) Nourish and remove waste
- B) Strengthen the joint
- C) Lubricate
- D) Remove debris
- E) Absorb shock

13. Which feature is UNIQUE to skeletal muscle fibres?

- A) Mitotic ability
- B) Caveolae
- C) Single central nucleus
- D) Branched fibres
- E) Triad at the A-I junction

14. Synchronous contraction of cardiomyocytes is enabled by:

- A) Gap junctions
- B) Tight junctions
- C) Hemidesmosomes
- D) Desmosomes
- E) Fascia adherens

15. Which organ contains NO smooth muscle?

- A) Uterus
- B) Stomach
- C) Artery
- D) Diaphragm
- E) Urinary bladder

16. The primary Ca^{2+} store in skeletal muscle is the:

- A) The T-tubule network

- B) The cell mitochondria
- C) The sarcoplasmic reticulum
- D) The cytosolic space
- E) The sarcolemma surface

17. At a cardiac Z-line you find a:

- A) Dense body
- B) Triad
- C) Caveola
- D) Diad
- E) Tetrad

18. A feature shared by skeletal and smooth muscle is:

- A) Satellite-cell repair
- B) Intercalated disc coupling
- C) Troponin-based regulation
- D) Visible cross-striations
- E) A calcium rise triggers it

19. Compared with type IIb/X fibres, type I fibres have higher:

- A) Contraction speed
- B) ATPase activity
- C) Fibre diameter
- D) Glycolytic enzymes
- E) Oxidative capacity

20. Which statement about skeletal muscle is correct?

- A) Caveolae carry the action potential
- B) Dense bodies anchor actin
- C) T-tubules are sarcolemma invaginations
- D) SR stores no calcium
- E) It regenerates by pericytes

21. A skin biopsy of the dermis most likely shows:

- A) Loose areolar tissue
- B) Adipose tissue
- C) Dense irregular connective tissue
- D) Dense regular connective tissue
- E) Reticular tissue

22. Which connective-tissue cell produces antibodies?

- A) Plasma cell
- B) Fibroblast
- C) Macrophage
- D) Eosinophil
- E) Mast cell

23. Weak neuromuscular transmission (as in myasthenia gravis) implicates the:

- A) Schwann cell
- B) Synaptic vesicles
- C) Motor end-plate
- D) Perimysium
- E) Axon terminal

24. Which is an example of dense regular connective tissue?

- A) Dermis
- B) Tendon
- C) Spleen capsule
- D) Adipose tissue
- E) Lymph node

- 25.** Botulinum toxin causes paralysis by acting at the:
- A) Sarcoplasmic reticulum pump
 - B) Motor end-plate junctional fold
 - C) Presynaptic ACh-release block
 - D) Postsynaptic ACh receptor
 - E) Voltage-gated Na channel
- 26.** Which statement about muscle cells is INCORRECT?
- A) Skeletal muscle has triads
 - B) Smooth muscle uses calmodulin
 - C) Type I fibres resist fatigue
 - D) Cardiac has intercalated discs
 - E) Smooth regenerates by satellite cells
- 27.** In smooth muscle, Ca^{2+} is associated with flask-shaped invaginations of the:
- A) T-tubule
 - B) Mitochondria
 - C) Nuclear envelope
 - D) Golgi
 - E) Sarcolemma (caveolae)
- 28.** Type I vs type IIb fibre classification is based on:
- A) Myosin-ATPase histochemistry
 - B) Total fibre length measured
 - C) Position of the nucleus
 - D) Sarcomere number per fibre
 - E) Capillary density counting
- 29.** Heavy resistance training chiefly causes:
- A) Hyperplasia of new fibres
 - B) Fibre-type switch to smooth
 - C) Satellite-cell depletion
 - D) Loss of connective tissue
 - E) Hypertrophy of myofibrils
- 30.** Prolonged limb casting causes muscle atrophy primarily via:
- A) Increased protein synthesis
 - B) Fat replacement of fibres
 - C) Calcium influx and necrosis
 - D) Satellite-cell proliferation
 - E) Increased protein degradation
- 31.** An example of isometric contraction is:
- A) Lifting a heavy dumbbell
 - B) Lowering a book slowly
 - C) Throwing a ball overhead
 - D) Squeezing glutes while still
 - E) Running fast downhill
- 32.** By Henneman's size principle, which motor units fire FIRST?
- A) Random units
 - B) Small, low-threshold
 - C) Large, high-threshold
 - D) Largest units
 - E) Fast-fatigable only
- 33.** Absence of dystrophin (Duchenne) primarily disrupts the:
- A) The cell nucleus core
 - B) The sarcoplasmic reticulum
 - C) The sarcolemma (costamere)

- D) The muscle mitochondria
- E) The Z-line of sarcomere

34. Which component is ABSENT in smooth-muscle cells?

- A) Calmodulin
- B) Myosin
- C) Dense bodies
- D) Actin
- E) Troponin C

35. Which tissue's matrix is richest in type II collagen?

- A) Dense tendon tissue
- B) Reticular dermis layer
- C) Hyaline cartilage matrix
- D) Elastic ligament tissue
- E) Compact bone matrix

36. Reticular fibres are classically demonstrated with:

- A) H&E
- B) PAS
- C) Oil red O
- D) Silver impregnation
- E) Masson trichrome

37. Glycogen in tissue is best detected histologically with:

- A) PAS reaction
- B) Immunohistochemistry
- C) Toluidine blue
- D) H&E
- E) Silver stain

38. The lymph-node framework (stroma) is composed of:

- A) Type I collagen bundles
- B) Reticular fibres
- C) Elastic fibres
- D) Smooth muscle
- E) Adipose tissue

39. Synovial joints differ from fibrous joints because they:

- A) Have a cavity, freely movable
- B) Are completely immovable
- C) Lack any joint cavity
- D) Have no fibrous capsule
- E) Contain red bone marrow

40. Laceration of a synovial (fascial) tendon sheath most likely injures the:

- A) Tendon
- B) Bursa
- C) Artery
- D) Nerve
- E) Bone

Section 2 · Cartilage, Bone & Integument (Lec 4–6)

41. Long bones grow in length through the:

- A) Endosteum
- B) Medullary cavity
- C) Bone collar
- D) Epiphyseal plate
- E) Periosteum

- 42.** The skin (integument) consists of which layers?
- A) Epidermis and dermis
 - B) Epidermis and hypodermis
 - C) Dermis and hypodermis
 - D) All three plus fascia
 - E) Epidermis only
- 43.** Hyaline-cartilage matrix contains which collagen?
- A) Type III
 - B) Type IV
 - C) Type II
 - D) Type I
 - E) Type V
- 44.** The basophilia of cartilage matrix is chiefly due to:
- A) Calcium salts
 - B) Elastic fibres
 - C) Proteoglycan sulphates
 - D) Water content
 - E) Type II collagen
- 45.** The main role of melanin pigment is to:
- A) Sense fine light touch
 - B) Protect against UV light
 - C) Lubricate the skin surface
 - D) Synthesise vitamin D fully
 - E) Insulate against heat loss
- 46.** A hypodermic injection FIRST pierces the:
- A) Epidermis
 - B) Hypodermis
 - C) Deep fascia
 - D) Dermis
 - E) Reticular dermis
- 47.** Which epidermal cell is part of the immune system?
- A) Keratinocyte
 - B) Merkel cell
 - C) Melanocyte
 - D) Basal cell
 - E) Langerhans cell
- 48.** The living 'prickle cells' of the epidermis are found in the stratum:
- A) Granulosum
 - B) Spinosum
 - C) Basale
 - D) Lucidum
 - E) Corneum
- 49.** Which structure is ABSENT in the stratum granulosum?
- A) Keratinocytes
 - B) Keratohyalin granules
 - C) Collagen fibres
 - D) Desmosomes
 - E) Lamellar bodies
- 50.** In the stratum corneum, conversion of LEKTI/KLK that drives desquamation depends mainly on:
- A) Oxygen
 - B) Temperature
 - C) Glucose

- D) pH
- E) Calcium only

51. Meissner corpuscles lie in the:

- A) Hypodermis
- B) Hair bulb
- C) Reticular dermis
- D) Stratum corneum
- E) Dermal papillae

52. Chondrocyte division within the matrix is called:

- A) Interstitial growth
- B) Endochondral growth
- C) Calcification
- D) Intramembranous growth
- E) Appositional growth

53. Intramembranous ossification begins within a:

- A) A stretched tendon body
- B) A vascular periosteal bud
- C) A hyaline cartilage model
- D) A pre-formed bony callus
- E) A fibrous mesenchymal sheet

54. The FIRST stage of fracture repair is:

- A) Bony callus
- B) Remodelling
- C) Soft callus
- D) Ossification
- E) Haematoma formation

55. Pathologic loss of bone MASS that compromises strength is:

- A) Osteomalacia
- B) Osteomyelitis
- C) Osteopetrosis
- D) Osteoarthritis
- E) Osteoporosis

56. Which feature does NOT describe an osteoclast?

- A) Resorbs bone
- B) Monocyte origin
- C) Ruffled border
- D) Multinucleated
- E) Secretes osteoid

57. Which statement about osteoblasts is FALSE?

- A) Of mesenchymal origin
- B) Secrete alkaline phosphatase
- C) Secrete osteoid
- D) Undergo mitosis
- E) Become osteocytes

58. The growth-plate zone where chondrocytes die is the zone of:

- A) Ossification
- B) Reserve
- C) Proliferation
- D) Hypertrophy
- E) Calcification

59. Which statement about cartilage is TRUE?

- A) Fibrocartilage's main collagen is Type I

- B) Articular cartilage is vascular
- C) Elastic cartilage lacks chondrocytes
- D) Fibrocartilage has a perichondrium
- E) Hyaline matrix is fibrous

60. The epidermal layer responsible for regeneration is the stratum:

- A) The stratum spinosum
- B) The stratum basale layer
- C) The stratum corneum layer
- D) The stratum granulosum
- E) The stratum lucidum band

61. Which is NOT an epidermal appendage?

- A) Arrector pili muscle
- B) Sweat gland
- C) Sebaceous gland
- D) Hair follicle
- E) Mammary gland

62. The bone collar is formed:

- A) At the secondary centre
- B) During adult remodelling
- C) During primary bone growth
- D) By the epiphyseal plate
- E) Within the marrow cavity

63. A feature typical of elastic cartilage is:

- A) Absence of perichondrium
- B) Elastic fibres in the matrix
- C) Calcified matrix
- D) Type I collagen only
- E) No chondrocytes

64. Articular discs and knee menisci are made chiefly of:

- A) Dense regular CT
- B) Elastic cartilage
- C) Fibrocartilage
- D) Bone
- E) Hyaline cartilage

65. The typical function of a chondroblast is to:

- A) Phagocytose debris
- B) Form bone
- C) Store fat
- D) Secrete cartilage matrix
- E) Resorb cartilage

66. Articular cartilage heals slowly chiefly because it:

- A) Its matrix is calcified
- B) It is richly innervated
- C) It contains many osteoclasts
- D) It is avascular, no perichondrium
- E) It is highly vascular tissue

67. During bone repair, osteoblasts first lay down osteoid in the stage of:

- A) Haematoma organisation stage
- B) Bone resorption stage
- C) Soft callus formation stage
- D) Final remodelling stage
- E) Bony (hard) callus stage

- 68.** Albinism results from:
- A) Excess eumelanin pigment
 - B) Loss of basal keratinocytes
 - C) Absence of all melanocytes
 - D) Destruction of basal layer
 - E) Lack of tyrosinase enzyme
- 69.** When body temperature rises, dermal vessels:
- A) Thrombose and occlude
 - B) Calcify within the dermis
 - C) Disappear into dermis
 - D) Dilate to raise skin flow
 - E) Constrict to retain heat
- 70.** A second-degree burn that loses fine touch but spares deeper structures most damages the:
- A) Ruffini ending
 - B) Free nerve ending
 - C) Merkel disc
 - D) Pacinian corpuscle
 - E) Meissner corpuscle
- 71.** The epidermal layer with both mitotic activity and pigment granules is the stratum:
- A) Corneum
 - B) Basale
 - C) Lucidum
 - D) Granulosum
 - E) Spinosum
- 72.** The territorial matrix of hyaline cartilage is:
- A) Proteoglycan-rich, collagen-poor
 - B) Vascular
 - C) Collagen-rich, proteoglycan-poor
 - D) Acellular
 - E) Calcified
- 73.** Hypertrichosis arises from abnormal development of the:
- A) Sweat gland
 - B) Hair follicle
 - C) Arrector pili
 - D) Mammary gland
 - E) Sebaceous gland

Section 3 · Back Muscles, Suboccipital Triangle & Vertebral Column (Lec 7–9)

- 74.** Which single muscle forms the inferior border of the triangle of auscultation AND the medial border of the lumbar triangle?
- A) External oblique
 - B) Trapezius
 - C) Latissimus dorsi
 - D) Rhomboid major
 - E) Serratus posterior inferior
- 75.** An abnormally increased thoracic curvature is called:
- A) Kyphosis
 - B) Spondylosis
 - C) Scoliosis
 - D) Lordosis
 - E) Subluxation
- 76.** Which structure contains ONLY sensory fibres?

- A) Spinal nerve
- B) Dorsal root
- C) White ramus
- D) Ventral root
- E) Grey ramus

77. Which structure contains NO sensory fibres?

- A) Ventral ramus
- B) Dorsal root
- C) Ventral root
- D) Dorsal ramus
- E) Spinal nerve

78. The trapezius receives motor innervation from the:

- A) Thoracodorsal nerve
- B) Long thoracic nerve
- C) Accessory nerve (CN XI)
- D) Dorsal scapular nerve
- E) C3–C4 only

79. Latissimus dorsi is innervated by the:

- A) Long thoracic nerve
- B) Thoracodorsal nerve
- C) Axillary nerve
- D) Accessory nerve
- E) Dorsal scapular nerve

80. The rhomboids and levator scapulae are supplied by the:

- A) Suprascapular nerve
- B) Dorsal scapular nerve
- C) Long thoracic nerve
- D) Thoracodorsal nerve
- E) Accessory nerve

81. Deep (intrinsic) back muscles such as erector spinae are innervated by the:

- A) Accessory nerve
- B) Dorsal primary rami
- C) Phrenic nerve
- D) Ventral primary rami
- E) Sympathetic chain

82. Upper cervical transverse-process fractures (C1–C4) could weaken the:

- A) Trapezius
- B) Rhomboid major
- C) Serratus anterior
- D) Latissimus dorsi
- E) Levator scapulae

83. Spondylolysis is most commonly a fracture/defect of the:

- A) Pars interarticularis
- B) Vertebral body
- C) Spinous process
- D) Pedicle
- E) Transverse process

84. The ROOF of the suboccipital triangle is formed by the:

- A) Obliquus capitis superior
- B) Rectus capitis posterior major
- C) Semispinalis capitis
- D) Obliquus capitis inferior

E) Trapezius

85. A hyperextension (whiplash) injury most often tears the:

- A) Posterior longitudinal ligament
- B) Ligamentum flavum
- C) Nuchal ligament
- D) Anterior longitudinal ligament
- E) Supraspinous ligament

86. The contents of the suboccipital triangle are the:

- A) Greater occipital nerve and vein
- B) Spinal accessory nerve and occipital artery
- C) Vertebral artery and suboccipital nerve
- D) Cervical plexus roots
- E) Internal carotid and C2

87. A stab wound to the spinal accessory nerve in the posterior triangle paralyses the:

- A) Serratus anterior
- B) Rhomboid major
- C) Levator scapulae
- D) Splenius capitis
- E) Trapezius

88. The suboccipital nerve (dorsal ramus of C1) spares which muscle?

- A) Splenius capitis
- B) Obliquus capitis superior
- C) Rectus capitis posterior minor
- D) Rectus capitis posterior major
- E) Obliquus capitis inferior

89. The lumbar (Petit) triangle is bounded by latissimus dorsi, iliac crest and the:

- A) Erector spinae
- B) Serratus posterior inferior
- C) Trapezius
- D) Quadratus lumborum
- E) External abdominal oblique

90. Erector spinae columns from lateral to medial are:

- A) Multifidus, rotatores, spinalis
- B) Longissimus, iliocostalis, spinalis
- C) Splenius, longissimus, spinalis
- D) Spinalis, longissimus, iliocostalis
- E) Iliocostalis, longissimus, spinalis

91. The transversospinalis group from superficial to deep is:

- A) Multifidus, rotatores, semispinalis
- B) Spinalis, longissimus, iliocostalis
- C) Rotatores, multifidus, semispinalis
- D) Splenius, semispinalis, rotatores
- E) Semispinalis, multifidus, rotatores

92. Rotation of the head ('NO' movement) occurs mainly at the:

- A) Atlanto-occipital joint
- B) C2–C3 disc
- C) Median atlanto-axial joint
- D) Zygapophyseal joints
- E) Costovertebral joint

93. The atlanto-occipital joint chiefly permits:

- A) Wide circumduction
- B) Pure rotation of head

- C) No movement at all
- D) Only lateral bending
- E) Flexion/extension nodding

94. Approximately 95% of lumbar disc herniations occur at:

- A) Levels L2 and L3
- B) Levels S1 and S2
- C) Levels L4–L5 or L5–S1
- D) Levels T12 and L1
- E) Levels L1 and L2

95. Rupture of the transverse ligament of the atlas leads to:

- A) C2–C3 disc herniation
- B) Simple loss of lordosis
- C) Atlanto-axial subluxation
- D) Pure hyperextension injury
- E) Bony fusion of C1–C2

96. The nucleus pulposus of the intervertebral disc is:

- A) The semifluid central core
- B) A fibrous outer ring
- C) A hyaline cartilage cap
- D) A synovial fluid pad
- E) An elastic ligament band

97. The ligament that limits flexion and is elastic, helping straighten the column, is the:

- A) Nuchal
- B) Supraspinous
- C) Tectorial membrane
- D) Ligamentum flavum
- E) Anterior longitudinal

98. Spondylolisthesis is best described as:

- A) Forward vertebral slippage
- B) A lateral spinal curve
- C) Inflammation of a vertebra
- D) Disc dehydration only
- E) A pars defect, no slip

Section 4 · Scapular, Deltoid, Pectoral & Axilla (Lec 10–11)

99. A fracture of the bony 'point' of the shoulder involves the:

- A) Coracoid process
- B) Clavicle shaft
- C) Greater tubercle
- D) Glenoid
- E) Acromion

100. Posterior-cord injury of the brachial plexus chiefly affects the:

- A) Median nerve
- B) Long thoracic nerve
- C) Musculocutaneous nerve
- D) Radial nerve
- E) Ulnar nerve

101. Which muscle initiates abduction of the arm (0–15°)?

- A) Subscapularis
- B) Teres minor
- C) Infraspinatus
- D) Deltoid

E) Supraspinatus

102. Upper-root (C5–C6, Erb) brachial-plexus injury spares the:

- A) Flexor carpi ulnaris
- B) Supraspinatus
- C) Deltoid
- D) Brachialis
- E) Biceps brachii

103. Excision of a mole just above the clavicle risks cutting the:

- A) Supraclavicular nerves
- B) Long thoracic nerve
- C) Vagus nerve
- D) Accessory nerve
- E) Phrenic nerve

104. Which axillary node group mainly drains the breast?

- A) The central node group
- B) The subscapular group
- C) The apical node group
- D) The pectoral (anterior) group
- E) The lateral node group

105. After modified mastectomy, weakness of the LOWER pectoralis major indicates injury to the:

- A) Long thoracic nerve
- B) Medial pectoral nerve
- C) Axillary nerve
- D) Thoracodorsal nerve
- E) Lateral pectoral nerve

106. Mid-shaft clavicle fracture with loss of elbow flexion suggests injury to the:

- A) The medial cord branch
- B) The posterior cord branch
- C) The lateral cord (musculocut.)
- D) The lower trunk branch
- E) The long thoracic nerve

107. A winged scapula (medial border lifts off) indicates injury to the:

- A) Thoracodorsal nerve
- B) Long thoracic nerve
- C) Axillary nerve
- D) Dorsal scapular nerve
- E) Accessory nerve

108. Anterior–inferior shoulder dislocation most endangers the:

- A) Axillary nerve
- B) Long thoracic nerve
- C) Musculocutaneous nerve
- D) Suprascapular nerve
- E) Radial nerve

109. Dorsal scapular-nerve injury weakens which muscle?

- A) Latissimus dorsi
- B) Trapezius
- C) Serratus anterior
- D) Deltoid
- E) Rhomboid major

110. The cephalic vein runs in the deltopectoral groove and drains into the:

- A) Axillary vein
- B) Basilic vein

- C) Brachial vein
- D) Subclavian vein directly
- E) Internal jugular vein

Section 5 · Arm, Forearm & Cubital Fossa (Lec 12–13)

- 111.** Which nerve lies posterior to the medial epicondyle of the humerus?
- A) Musculocutaneous nerve
 - B) Radial nerve
 - C) Median nerve
 - D) Axillary nerve
 - E) Ulnar nerve
- 112.** The lateral cutaneous nerve of the forearm is the continuation of the:
- A) The musculocutaneous nerve
 - B) The radial nerve branch
 - C) The ulnar nerve trunk
 - D) The axillary nerve branch
 - E) The median nerve trunk
- 113.** An absent biceps reflex indicates a lesion at which spinal level?
- A) C8
 - B) C7
 - C) C5
 - D) T1
 - E) L4
- 114.** Which muscle medially rotates the arm?
- A) Deltoid (posterior)
 - B) Teres minor
 - C) Latissimus dorsi
 - D) Supraspinatus
 - E) Infraspinatus
- 115.** Which cutaneous nerve of the upper limb is NOT from the posterior cord?
- A) Upper lateral cutaneous of arm
 - B) Lower lateral cutaneous of arm
 - C) Lateral cutaneous of forearm
 - D) Posterior cutaneous of arm
 - E) Posterior cutaneous of forearm
- 116.** Fracture of which humeral part may injure the ulnar nerve?
- A) Medial epicondyle
 - B) Surgical neck
 - C) Lateral epicondyle
 - D) Capitulum
 - E) Mid-shaft
- 117.** A high median-nerve injury at the elbow chiefly causes:
- A) Loss of arm abduction
 - B) Pure wrist-drop deformity
 - C) Claw of all the fingers
 - D) Loss of elbow extension
 - E) Weak forearm flexion/pronation
- 118.** Among brachial-plexus branches, which supplies an EXTENSOR of the arm?
- A) Thoracodorsal
 - B) Upper subscapular
 - C) Long thoracic
 - D) Musculocutaneous

E) Medial pectoral

119. Olecranon fracture causes weakness of elbow extension because it detaches the:

- A) Supinator
- B) Triceps insertion
- C) Biceps
- D) Brachialis
- E) Brachioradialis

120. In the cubital fossa, the order from superficial to deep is:

- A) Skin, vein, aponeurosis, artery
- B) Brachial artery then vein
- C) Artery, nerve, vein, skin
- D) Median nerve, then artery
- E) Aponeurosis, skin, artery

121. A lateral-epicondyle (common extensor origin) fracture SPARES which muscle?

- A) Extensor digiti minimi
- B) Extensor indicis
- C) Extensor digitorum
- D) Extensor carpi radialis brevis
- E) Extensor carpi ulnaris

122. For surgery at the SURGICAL NECK of the humerus, the preferred block is:

- A) Digital block
- B) Axillary block
- C) Bier block
- D) Wrist block
- E) Interscalene block

123. For mid-shaft humerus and distal surgery, a suitable block is the:

- A) Interscalene
- B) Stellate ganglion block
- C) Axillary block
- D) Phrenic block
- E) Cervical plexus block

124. Klumpke palsy (lower trunk C8–T1) classically produces:

- A) Ulnar claw hand
- B) Winged scapula
- C) Deltoid wasting
- D) Wrist drop
- E) Waiter's-tip posture

125. Pain on resisted elbow flexion in SUPINATION points to injury of the:

- A) Pronator teres
- B) Coracobrachialis
- C) Brachialis
- D) Brachioradialis
- E) Biceps brachii

126. Inability to supinate the forearm is most often accompanied by:

- A) Loss of pronation
- B) Loss of thumb opposition
- C) Weak elbow extension
- D) Weak finger flexion
- E) Loss of arm abduction

127. Which forearm muscle–nerve pairing is CORRECT?

- A) FCU — median nerve supply
- B) Pronator teres — ulnar

- C) FPL — ulnar nerve supply
- D) EI — posterior interosseous
- E) Brachioradialis — median

128. A spiral (mid-shaft) humeral fracture classically causes:

- A) Ape hand
- B) Waiter's tip
- C) Winged scapula
- D) Wrist drop
- E) Claw hand

129. 'Golfer's elbow' (medial epicondylitis) involves the:

- A) Olecranon bursa
- B) Biceps tendon
- C) Common flexor origin
- D) Common extensor origin
- E) Radial head

130. 'Saturday-night palsy' produces:

- A) Claw hand
- B) Loss of thumb opposition
- C) Loss of abduction
- D) Loss of pronation
- E) Wrist drop

131. A Colles' fracture is a distal radius fracture with:

- A) Ulnar shaft fracture only
- B) Radial head dislocation
- C) No displacement
- D) Posterior (dorsal) displacement
- E) Anterior (volar) displacement

132. The brachial artery divides into radial and ulnar arteries at the:

- A) Carpal tunnel
- B) Apex of the cubital fossa
- C) Mid-arm
- D) Deltopectoral triangle
- E) Wrist

Section 6 · Hand & Joints of Upper Limb (Lec 14–15)

133. The radio-ulnar interosseous membrane is a:

- A) Syndesmosis
- B) Suture
- C) Synchondrosis
- D) Symphysis
- E) Gomphosis

134. Which ligament reinforces the sternoclavicular joint and limits girdle elevation?

- A) Coraco-acromial
- B) Coracoclavicular
- C) Acromioclavicular
- D) Costoclavicular ligament
- E) Conoid

135. A wrist injury gives weak thumb flexion, abduction and inability to oppose, with NO sensory loss. The injured structure is the:

- A) Ulnar nerve at Guyon's
- B) Deep ulnar branch
- C) Median nerve proper

- D) Recurrent branch of median nerve
- E) Anterior interosseous nerve

136. A claw-hand deformity signals injury to the:

- A) Axillary nerve
- B) Musculocutaneous nerve
- C) Ulnar nerve
- D) Radial nerve
- E) Median nerve

137. Anterior shoulder dislocation does NOT injure which structure (it stabilises the AC, not GH, joint)?

- A) Coracoclavicular ligament
- B) Posterior circumflex artery
- C) Glenoid labrum
- D) Joint capsule
- E) Axillary nerve

138. The carpal tunnel transmits the median nerve and:

- A) Extensor tendons
- B) Flexor tendons
- C) Ulnar artery
- D) Radial artery
- E) Ulnar nerve

139. On the ulnar side, the flexor retinaculum attaches to the:

- A) Pisiform and hook of hamate
- B) Lunate and capitate
- C) Bases of metacarpals
- D) Radius styloid
- E) Scaphoid and trapezium

140. Most intrinsic hand muscles are supplied by the deep branch of the ulnar nerve EXCEPT the thenar muscles and the:

- A) Hypothenar muscles
- B) Adductor pollicis
- C) Lateral two lumbricals
- D) All interossei
- E) Palmaris brevis

141. The prime mover of thumb opposition is the:

- A) Extensor pollicis longus
- B) Abductor pollicis brevis
- C) Flexor pollicis brevis
- D) Adductor pollicis
- E) Opponens pollicis

142. The dorsal interossei of the hand act to:

- A) Adduct the fingers
- B) Extend the wrist
- C) Flex the wrist
- D) Oppose the thumb
- E) Abduct the fingers

143. The lumbricals flex the MCP joints and:

- A) Abduct the fingers
- B) Extend the IP joints
- C) Flex the IP joints
- D) Oppose the little finger
- E) Adduct the thumb

144. The elbow (humero-ulnar) joint is a:

- A) Saddle joint
- B) Plane joint
- C) Ball-and-socket
- D) Pivot joint
- E) Hinge joint

145. The wrist (radiocarpal) joint is a:

- A) A flat plane joint
- B) A uniaxial pivot joint
- C) A biaxial saddle joint
- D) A condyloid (ellipsoid) joint
- E) A simple hinge joint

146. The proximal radio-ulnar joint is a:

- A) Condyloid joint
- B) Saddle joint
- C) Hinge joint
- D) Pivot joint
- E) Plane joint

147. The 1st carpometacarpal (thumb) joint is a:

- A) Saddle joint
- B) Ball-and-socket
- C) Plane joint
- D) Pivot joint
- E) Hinge joint

148. The glenoid cavity is deepened by the:

- A) Coracoacromial ligament
- B) Subacromial bursa
- C) Rotator interval
- D) Transverse humeral ligament
- E) Glenoid labrum

149. The four rotator cuff muscles are supraspinatus, infraspinatus, teres minor and:

- A) Deltoid
- B) Teres major
- C) Subscapularis
- D) Latissimus dorsi
- E) Coracobrachialis

150. 'Nursemaid's elbow' (radial-head subluxation) involves the:

- A) Quadratus ligament
- B) Radial collateral ligament
- C) Oblique cord
- D) Annular ligament
- E) Interosseous membrane

151. The chief flexor and chief extensor of the elbow are, respectively:

- A) Pronator teres; biceps
- B) Brachioradialis; supinator
- C) Brachialis; triceps brachii
- D) Biceps; anconeus
- E) Coracobrachialis; triceps

152. The anatomical snuffbox floor is formed by the scaphoid and:

- A) Pisiform
- B) Hamate
- C) Trapezium
- D) Capitate

E) Lunate

Section 7 · Development of Bone, Skin & Muscle (Lec 16–17)

153. Premature fusion of the SAGITTAL suture produces a long, narrow skull termed:

- A) Trigonoccephaly
- B) Scaphocephaly
- C) Brachycephaly
- D) Oxycephaly
- E) Plagiocephaly

154. Premature bilateral fusion of the CORONAL sutures gives a short, broad skull termed:

- A) Plagiocephaly
- B) Scaphocephaly
- C) Oxycephaly
- D) Trigonoccephaly
- E) Brachycephaly

155. Vertebrae and ribs develop from the:

- A) Neural crest
- B) Myotome
- C) Dermatome
- D) Sclerotome
- E) Lateral plate mesoderm

156. Skeletal muscle of the limbs develops from the:

- A) Sclerotome
- B) Dermatome
- C) Endoderm
- D) Neural crest
- E) Myotome

157. The flat bones of the skull vault form by:

- A) Appositional cartilage growth
- B) Fibrocartilage conversion
- C) Interstitial growth
- D) Intramembranous ossification
- E) Endochondral ossification

158. The skull base (sphenoid, ethmoid, occipital) forms by:

- A) Direct mesenchymal bone
- B) Intramembranous ossification
- C) Endochondral ossification
- D) No ossification
- E) Fibrous union

159. Congenital absence of the sternocostal head of pectoralis major with ipsilateral hand defects is:

- A) Prune-belly syndrome
- B) VACTERL
- C) Poland syndrome
- D) Klippel–Feil syndrome
- E) Sprengel deformity

160. Partial/complete congenital absence of abdominal-wall musculature characterises:

- A) Omphalocele
- B) Prune-belly syndrome
- C) Diastasis recti
- D) Gastroschisis
- E) Poland syndrome

161. Fusion/webbing of adjacent digits is termed:

- A) Brachydactyly
- B) Ectrodactyly
- C) Polydactyly
- D) Clinodactyly
- E) Syndactyly

162. Bilateral clubfoot (talipes equinovarus) centres on a deformity involving the:

- A) Metatarsals
- B) Cuboid
- C) Talus
- D) Navicular
- E) Calcaneus

Section 8 · Gluteal Region, Posterior Thigh & Popliteal Fossa (Lec 18–19)

163. The safe quadrant for a deep IM gluteal injection (avoiding the sciatic nerve) is the:

- A) Inferolateral
- B) Central
- C) Superolateral
- D) Superomedial
- E) Inferomedial

164. Gluteus maximus is innervated by the:

- A) Femoral nerve
- B) Superior gluteal nerve
- C) Inferior gluteal nerve
- D) Pudendal nerve
- E) Sciatic nerve

165. Gluteus medius and minimus are supplied by the:

- A) Femoral nerve
- B) Superior gluteal nerve
- C) Obturator nerve
- D) Inferior gluteal nerve
- E) Sciatic nerve

166. Paralysis of gluteus medius/minimus produces a positive:

- A) Babinski sign
- B) Lasegue sign
- C) Romberg sign
- D) Phalen sign
- E) Trendelenburg sign

167. Gluteus maximus chiefly produces which action at the hip?

- A) Hip flexion only
- B) Extension, lateral rotation
- C) Pure hip flexion
- D) Internal hip rotation
- E) Strong hip adduction

168. Gluteus medius and minimus together produce abduction and:

- A) Strong hip adduction
- B) Medial thigh rotation
- C) Pure hip extension
- D) Lateral thigh rotation
- E) Hip flexion alone

169. A posterior hip dislocation most often injures the:

- A) Superior gluteal nerve

- B) Femoral nerve
- C) Sciatic nerve
- D) Obturator nerve
- E) Pudendal nerve

170. Avascular necrosis of the femoral head most often follows damage to the:

- A) Medial circumflex femoral artery
- B) Inferior gluteal artery
- C) Lateral circumflex femoral artery
- D) Profunda femoris artery
- E) Obturator artery

171. In the popliteal fossa, the MOST superficial central structure is the:

- A) Popliteal vein
- B) Small saphenous vein
- C) Popliteal artery
- D) Common fibular nerve
- E) Tibial nerve

172. Which muscle is NOT a 'true hamstring' (supplied by the common fibular division)?

- A) Semitendinosus
- B) Long head of biceps femoris
- C) All are true hamstrings
- D) Short head of biceps femoris
- E) Semimembranosus

173. Which structure exits the greater sciatic foramen ABOVE piriformis?

- A) Sciatic nerve
- B) Inferior gluteal nerve
- C) Pudendal nerve
- D) Posterior femoral cutaneous nerve
- E) Superior gluteal nerve

174. The gluteal ligaments that convert the sciatic notches into foramina are the sacrospinous and:

- A) The iliolumbar ligament
- B) The inguinal ligament
- C) The iliofemoral ligament
- D) The sacrotuberous ligament
- E) The pubofemoral ligament

Section 9 · Anterior/Medial Thigh, Leg, Foot & Lower-limb Joints (Lec 20–23)

175. The anterior compartment of the thigh (quadriceps) is innervated by the:

- A) Femoral nerve
- B) Sciatic nerve
- C) Obturator nerve
- D) Tibial nerve
- E) Superior gluteal nerve

176. The medial (adductor) compartment of the thigh is innervated by the:

- A) Sciatic nerve
- B) Common fibular nerve
- C) Tibial nerve
- D) Femoral nerve
- E) Obturator nerve

177. The anterior compartment of the leg is supplied by the:

- A) Tibial nerve
- B) Superficial fibular nerve
- C) Sural nerve

- D) Deep fibular nerve
- E) Saphenous nerve

178. The lateral compartment of the leg is supplied by the:

- A) Superficial fibular nerve
- B) Tibial nerve
- C) Deep fibular nerve
- D) Saphenous nerve
- E) Sural nerve

179. The posterior compartment of the leg is supplied by the:

- A) Tibial nerve
- B) Saphenous nerve
- C) Sural nerve
- D) Deep fibular nerve
- E) Superficial fibular nerve

180. The main action of the anterior leg compartment is:

- A) Dorsiflexion
- B) Inversion
- C) Eversion
- D) Plantarflexion
- E) Toe flexion

181. The lateral leg compartment produces:

- A) Eversion
- B) Dorsiflexion
- C) Plantarflexion only
- D) Toe extension
- E) Inversion

182. The triceps surae (gastrocnemius + soleus) act to:

- A) Extend the toes
- B) Evert
- C) Plantarflex the ankle
- D) Invert
- E) Dorsiflex

183. The muscle that 'unlocks' the fully extended knee is the:

- A) Soleus
- B) Tibialis posterior
- C) Gracilis
- D) Popliteus
- E) Sartorius

184. A fracture at the neck of the fibula most commonly injures the:

- A) Sural nerve
- B) Femoral nerve
- C) Common fibular nerve
- D) Tibial nerve
- E) Saphenous nerve

185. Foot drop (loss of dorsiflexion and eversion) follows injury to the:

- A) Common fibular nerve
- B) Obturator nerve
- C) Tibial nerve
- D) Femoral nerve
- E) Saphenous nerve

186. Tarsal-tunnel syndrome compresses the:

- A) Sural nerve

- B) Deep fibular nerve
- C) Saphenous nerve
- D) Tibial nerve
- E) Common fibular nerve

187. Tibialis posterior chiefly produces:

- A) Dorsiflexion and eversion
- B) Toe extension
- C) Plantarflexion and inversion
- D) Knee flexion
- E) Eversion only

188. Collapse of the medial longitudinal arch (flatfoot) classically follows dysfunction of the:

- A) Fibularis longus
- B) Flexor hallucis longus
- C) Gastrocnemius
- D) Tibialis posterior tendon
- E) Tibialis anterior

189. The hip joint is which type of synovial joint?

- A) Pivot
- B) Ball-and-socket
- C) Hinge
- D) Condylloid
- E) Saddle

190. The strongest hip ligament, limiting hyperextension, is the:

- A) Iliofemoral ligament
- B) Pubofemoral ligament
- C) Ischiofemoral ligament
- D) Transverse acetabular
- E) Ligament of the head

191. The knee is best described as a:

- A) Pivot
- B) Ball-and-socket
- C) Plane joint
- D) Modified hinge joint
- E) Simple hinge

192. The anterior cruciate ligament arises from which area of the tibia?

- A) Tibial tuberosity
- B) Anterior intercondylar area
- C) Medial condyle
- D) Posterior intercondylar area
- E) Lateral meniscus

193. The posterior cruciate ligament prevents:

- A) Posterior tibial displacement
- B) Hyperextension only
- C) Valgus angulation
- D) Varus angulation
- E) Anterior tibial displacement

194. Which meniscus is C-shaped and firmly attached to the tibial collateral ligament?

- A) Neither
- B) The transverse ligament
- C) Both equally
- D) Lateral meniscus
- E) Medial meniscus

- 195.** The acetabulum is formed by ilium, ischium and:
- A) Sacrum
 - B) Sacrotuberous ligament
 - C) Femur
 - D) Coccyx
 - E) Pubis
- 196.** The great saphenous vein ultimately drains into the:
- A) External iliac vein
 - B) Femoral vein
 - C) Small saphenous vein
 - D) Dorsal venous arch
 - E) Popliteal vein
- 197.** The small saphenous vein drains into the:
- A) Femoral vein
 - B) External iliac vein
 - C) Great saphenous vein
 - D) Popliteal vein
 - E) Dorsal venous arch
- 198.** At heel-strike, the ankle is held in:
- A) Inversion
 - B) Neutral with toe flexion
 - C) Eversion
 - D) Plantarflexion
 - E) Dorsiflexion
- 199.** The popliteal artery divides into the anterior tibial and:
- A) Medial plantar artery
 - B) Femoral artery
 - C) Posterior tibial artery
 - D) Dorsalis pedis
 - E) Fibular artery
- 200.** Collapse-resistant 'close-packed' stability of the ankle occurs in:
- A) Neutral
 - B) Plantarflexion
 - C) Inversion
 - D) Dorsiflexion
 - E) Eversion
- 201.** The hindfoot consists of the talus and:
- A) Metatarsals
 - B) Cuboid
 - C) Calcaneus
 - D) Cuneiforms
 - E) Navicular
- 202.** The plantar aponeurosis arises posteriorly from the:
- A) Talus
 - B) Calcaneus
 - C) Cuboid
 - D) First metatarsal
 - E) Navicular

Answer Key & Explanations

Concise reasoning for every item.

Section 1 · Anatomical Terms, Connective & Muscular Tissue (Lec 1–3)

1. Which structure is NOT part of the neuromuscular junction?

Answer: E) Intrafusal fibre

Intrafusal fibres belong to the muscle spindle (a sensory proprioceptor), not the motor NMJ. The NMJ comprises the axon terminal, synaptic cleft and motor end-plate (with junctional folds).

2. H&E staining works mainly on the principle of:

Answer: B) Cationic–anionic affinity

H&E depends on acid–base (ionic) chemistry: basophilic structures bind cationic haematoxylin; acidophilic structures bind anionic eosin.

3. Passing fixed tissue through graded ethanol is the step termed:

Answer: E) Dehydration

Dehydration removes water using ascending ethanol grades (70→90→100%) before clearing and paraffin infiltration.

4. With toluidine blue, which cell component is MOST basophilic?

Answer: D) Nucleolus

The nucleolus is rich in rRNA (highly acidic from phosphate groups), so it binds basic dyes most intensely.

5. Which connective-tissue cell shows an eccentric 'clock-face' nucleus?

Answer: B) Plasma cell

Plasma cells have an eccentric nucleus with peripheral heterochromatin clumps resembling a clock face; their basophilic cytoplasm is rich in rough ER for antibody synthesis.

6. Early-wound fibres that are silver- and PAS-positive are:

Answer: A) Collagen III

Type III collagen forms reticular fibres, prominent in granulation tissue; they are argyrophilic (silver+) and PAS-positive.

7. Which collagen type builds the basal lamina?

Answer: A) Type IV

Type IV collagen forms the mesh-like network of the basal lamina.

8. In elastic arteries, elastin of the tunica media is secreted by:

Answer: D) Smooth-muscle cells

In large-vessel walls, smooth-muscle cells produce the matrix, including collagen and elastin.

9. Which polyanion lets ground substance form a hydrated gel?

Answer: A) Hyaluronic acid

Hyaluronic acid is a very large, non-sulfated GAG that binds water to form the hydrated gel of ground substance.

10. The classic model organ for reticular tissue is the:

Answer: A) Spleen

The stroma of lymphoid/haematopoietic organs such as spleen and lymph nodes is reticular tissue (Type III collagen).

11. Articular cartilage and bursae are typical of which joint type?

Answer: A) Synovial (knee) joint

These are features of a synovial joint (joint cavity, synovial fluid, hyaline articular cartilage, often bursae/menisci).

12. Which is NOT a function of synovial fluid?

Answer: B) Strengthen the joint

Joint strength comes from the fibrous capsule and ligaments; synovial fluid lubricates, absorbs shock and exchanges nutrients/waste for avascular cartilage.

13. Which feature is UNIQUE to skeletal muscle fibres?

Answer: E) Triad at the A–I junction

A triad (one T-tubule + two SR cisternae) at the A–I junction is the defining feature of skeletal muscle; cardiac has diads at the Z-line.

14. Synchronous contraction of cardiomyocytes is enabled by:

Answer: A) Gap junctions

Gap junctions in intercalated discs electrically couple cardiac cells, allowing the heart to act as a syncytium.

15. Which organ contains NO smooth muscle?

Answer: D) Diaphragm

The diaphragm is entirely skeletal muscle; the others have smooth muscle in their walls.

16. The primary Ca^{2+} store in skeletal muscle is the:

Answer: C) The sarcoplasmic reticulum

The sarcoplasmic reticulum sequesters and releases Ca^{2+} to regulate contraction.

17. At a cardiac Z-line you find a:

Answer: D) Diad

Cardiac muscle has diads (one T-tubule + one SR cisterna) at the Z-line.

18. A feature shared by skeletal and smooth muscle is:

Answer: E) A calcium rise triggers it

Although regulatory proteins differ (troponin vs calmodulin), a rise in intracellular Ca^{2+} is the universal contraction trigger.

19. Compared with type IIb/X fibres, type I fibres have higher:

Answer: E) Oxidative capacity

Type I (slow) fibres are aerobic/endurance fibres with more mitochondria, myoglobin and capillaries — high oxidative capacity.

20. Which statement about skeletal muscle is correct?

Answer: C) T-tubules are sarcolemma invaginations

T-tubules are deep invaginations of the sarcolemma that conduct the action potential into the fibre interior.

21. A skin biopsy of the dermis most likely shows:

Answer: C) Dense irregular connective tissue

The reticular dermis is dense irregular connective tissue (interwoven collagen for multidirectional strength).

22. Which connective-tissue cell produces antibodies?

Answer: A) Plasma cell

Plasma cells are differentiated B-lymphocytes that synthesise and secrete immunoglobulins.

23. Weak neuromuscular transmission (as in myasthenia gravis) implicates the:

Answer: C) Motor end-plate

The motor end-plate is the postsynaptic membrane; loss of its ACh receptors (myasthenia gravis) weakens transmission.

24. Which is an example of dense regular connective tissue?

Answer: B) Tendon

Tendon is densely packed parallel Type I collagen — dense regular CT. Dermis and capsules are dense irregular.

25. Botulinum toxin causes paralysis by acting at the:

Answer: C) Presynaptic ACh-release block

Botulinum toxin cleaves SNARE proteins presynaptically, preventing ACh vesicle release.

26. Which statement about muscle cells is INCORRECT?

Answer: E) Smooth regenerates by satellite cells

Satellite cells are specific to skeletal muscle. Smooth muscle regenerates by mitosis of existing cells or from pericytes.

27. In smooth muscle, Ca^{2+} is associated with flask-shaped invaginations of the:

Answer: E) Sarcolemma (caveolae)

Caveolae are small invaginations of the sarcolemma in smooth muscle involved in calcium signalling (smooth muscle lacks T-tubules).

28. Type I vs type IIb fibre classification is based on:

Answer: A) Myosin-ATPase histochemistry

Fibres are classified by myosin-ATPase activity; fast (II) have high activity, slow (I) low activity.

29. Heavy resistance training chiefly causes:

Answer: E) Hypertrophy of myofibrils

Resistance training enlarges existing fibres by adding myofibrils (hypertrophy), not by adding fibres (hyperplasia).

30. Prolonged limb casting causes muscle atrophy primarily via:

Answer: E) Increased protein degradation

Disuse atrophy is driven by upregulated protein breakdown (ubiquitin-proteasome) exceeding synthesis.

31. An example of isometric contraction is:

Answer: D) Squeezing glutes while still

Isometric contraction develops tension without length change — squeezing the glutes while standing still. The others involve movement (isotonic).

32. By Henneman's size principle, which motor units fire FIRST?

Answer: B) Small, low-threshold

The smallest, lowest-threshold (Type I) units are recruited first; larger, more powerful units are added as force demand rises.

33. Absence of dystrophin (Duchenne) primarily disrupts the:

Answer: C) The sarcolemma (costamere)

Dystrophin links the actin cytoskeleton to the sarcolemma/ECM (costamere); its loss destabilises the sarcolemma, causing degeneration.

34. Which component is ABSENT in smooth-muscle cells?

Answer: E) Troponin C

Smooth muscle lacks troponin; Ca^{2+} binds calmodulin, which activates myosin light-chain kinase (MLCK).

35. Which tissue's matrix is richest in type II collagen?

Answer: C) Hyaline cartilage matrix

Type II collagen is the hallmark of hyaline cartilage; bone/tendon/dermis are mainly Type I.

36. Reticular fibres are classically demonstrated with:

Answer: D) Silver impregnation

Reticular fibres (Type III) are argyrophilic and stain black with silver impregnation.

37. Glycogen in tissue is best detected histologically with:

Answer: A) PAS reaction

The Periodic acid–Schiff (PAS) reaction stains complex carbohydrates, including glycogen, magenta.

38. The lymph-node framework (stroma) is composed of:

Answer: B) Reticular fibres

Lymphoid stroma is a meshwork of reticular fibres (Type III collagen).

39. Synovial joints differ from fibrous joints because they:

Answer: A) Have a cavity, freely movable

A fluid-filled joint cavity allowing free movement (diarthrosis) defines a synovial joint; fibrous joints lack a cavity.

40. Laceration of a synovial (fascial) tendon sheath most likely injures the:

Answer: A) Tendon

A synovial tendon sheath surrounds a tendon to reduce friction; lacerating it implicates the enclosed tendon.

Section 2 · Cartilage, Bone & Integument (Lec 4–6)

41. Long bones grow in length through the:

Answer: D) Epiphyseal plate

The epiphyseal (growth) plate is hyaline cartilage where interstitial growth lengthens the bone.

42. The skin (integument) consists of which layers?

Answer: A) Epidermis and dermis

The skin proper is epidermis + dermis; the hypodermis is subcutaneous tissue, not part of the skin itself.

43. Hyaline-cartilage matrix contains which collagen?

Answer: C) Type II

Hyaline cartilage matrix has fine Type II collagen with proteoglycans and hyaluronic acid.

44. The basophilia of cartilage matrix is chiefly due to:

Answer: C) Proteoglycan sulphates

Sulfated GAGs of proteoglycans are negatively charged (acidic), so they bind basic dyes — basophilia.

45. The main role of melanin pigment is to:

Answer: B) Protect against UV light

Melanin absorbs UV radiation, protecting keratinocyte DNA in the basal layer.

46. A hypodermic injection FIRST pierces the:

Answer: A) Epidermis

The needle first passes the outermost layer, the epidermis.

47. Which epidermal cell is part of the immune system?

Answer: E) Langerhans cell

Langerhans cells are dendritic antigen-presenting cells of the skin's immune defence.

48. The living 'prickle cells' of the epidermis are found in the stratum:

Answer: B) Spinosum

Keratinocytes of the stratum spinosum appear spiny ('prickle cells') because of prominent desmosomes.

49. Which structure is ABSENT in the stratum granulosum?

Answer: C) Collagen fibres

Collagen belongs to the dermis (connective tissue), not the avascular cellular epidermis.

50. In the stratum corneum, conversion of LEKTI/KLK that drives desquamation depends mainly on:

Answer: D) pH

The acidic pH of the stratum corneum governs the protease activity (KLK) that breaks corneodesmosomes for shedding.

51. Meissner corpuscles lie in the:

Answer: E) Dermal papillae

Meissner corpuscles (light touch) sit superficially in the dermal papillae.

52. Chondrocyte division within the matrix is called:

Answer: A) Interstitial growth

Interstitial growth is division of existing chondrocytes within the matrix; appositional growth adds cells at the surface.

53. Intramembranous ossification begins within a:

Answer: E) A fibrous mesenchymal sheet

Intramembranous ossification forms bone directly within a mesenchymal/fibrous membrane (flat skull bones).

54. The FIRST stage of fracture repair is:

Answer: E) Haematoma formation

A fracture haematoma (clot) forms first from ruptured vessels.

55. Pathologic loss of bone MASS that compromises strength is:

Answer: E) Osteoporosis

Osteoporosis = low bone mass and microarchitectural deterioration, raising fracture risk. Osteomalacia is defective mineralisation.

56. Which feature does NOT describe an osteoclast?

Answer: E) Secretes osteoid

Osteoclasts resorb bone; osteoblasts secrete osteoid.

57. Which statement about osteoblasts is FALSE?

Answer: D) Undergo mitosis

Osteoblasts are terminally differentiated and do not divide; their osteoprogenitors do.

58. The growth-plate zone where chondrocytes die is the zone of:

Answer: E) Calcification

In the calcification zone the matrix mineralises, cutting off nutrients so hypertrophied chondrocytes die.

59. Which statement about cartilage is TRUE?

Answer: A) Fibrocartilage's main collagen is Type I

Fibrocartilage is rich in Type I collagen (with some II); it lacks a perichondrium, unlike hyaline/elastic cartilage.

60. The epidermal layer responsible for regeneration is the stratum:

Answer: B) The stratum basale layer

The stratum basale contains mitotically active stem cells that regenerate the epidermis.

61. Which is NOT an epidermal appendage?

Answer: A) Arrector pili muscle

The arrector pili is smooth muscle of mesenchymal origin; the glands and hair follicle are epidermal derivatives.

62. The bone collar is formed:

Answer: C) During primary bone growth

In endochondral ossification the periosteal bone collar is the first bone, around the diaphysis of the cartilage model.

63. A feature typical of elastic cartilage is:

Answer: B) Elastic fibres in the matrix

Elastic cartilage resembles hyaline but has abundant elastic fibres giving flexibility (e.g., external ear, epiglottis).

64. Articular discs and knee menisci are made chiefly of:

Answer: C) Fibrocartilage

Discs and menisci are fibrocartilage (with a hyaline component), suited to shock absorption and tensile load.

65. The typical function of a chondroblast is to:

Answer: D) Secrete cartilage matrix

Chondroblasts secrete cartilage matrix; once surrounded by it they become chondrocytes.

66. Articular cartilage heals slowly chiefly because it:

Answer: D) It is avascular, no perichondrium

Lacking a perichondrium (no progenitor source) and being avascular (diffusion only) severely limits its repair.

67. During bone repair, osteoblasts first lay down osteoid in the stage of:

Answer: E) Bony (hard) callus stage

After the soft (fibrocartilaginous) callus, osteoblasts deposit osteoid that mineralises into the bony (hard) callus.

68. Albinism results from:

Answer: E) Lack of tyrosinase enzyme

Melanocytes are present but lack tyrosinase, so melanin cannot be made.

69. When body temperature rises, dermal vessels:

Answer: D) Dilate to raise skin flow

Vasodilation brings warm blood to the surface to lose heat by radiation/convection.

70. A second-degree burn that loses fine touch but spares deeper structures most damages the:

Answer: E) Meissner corpuscle

Meissner corpuscles (fine touch) lie superficially in dermal papillae and are lost first; deeper Pacinian/Ruffini are spared.

71. The epidermal layer with both mitotic activity and pigment granules is the stratum:

Answer: B) Basale

The stratum basale is mitotically active and houses melanocytes that transfer melanin to keratinocytes.

72. The territorial matrix of hyaline cartilage is:

Answer: A) Proteoglycan-rich, collagen-poor

The territorial matrix immediately around chondrocytes is intensely basophilic (sulfated proteoglycan-rich) and relatively collagen-poor.

73. Hypertrichosis arises from abnormal development of the:

Answer: B) Hair follicle

Hypertrichosis (excess hair) reflects abnormal hair-follicle development/activity.

Section 3 · Back Muscles, Suboccipital Triangle & Vertebral Column (Lec 7–9)

74. Which single muscle forms the inferior border of the triangle of auscultation AND the medial border of the lumbar triangle?

Answer: C) Latissimus dorsi

Latissimus dorsi borders both: inferior border of the auscultation triangle and medial border of the lumbar (Petit) triangle.

75. An abnormally increased thoracic curvature is called:

Answer: A) Kyphosis

Kyphosis is excessive posterior thoracic curvature; lordosis is excessive lumbar/cervical, scoliosis is lateral.

76. Which structure contains ONLY sensory fibres?

Answer: B) Dorsal root

The dorsal root is purely sensory (cell bodies in the dorsal root ganglion).

77. Which structure contains NO sensory fibres?

Answer: C) Ventral root

The ventral root is purely motor (efferent).

78. The trapezius receives motor innervation from the:

Answer: C) Accessory nerve (CN XI)

Trapezius motor supply is the spinal accessory nerve (CN XI); C3–C4 give sensory (proprioceptive) fibres.

79. Latissimus dorsi is innervated by the:

Answer: B) Thoracodorsal nerve

The thoracodorsal nerve (C6–C8, posterior cord) is the sole motor nerve to latissimus dorsi.

80. The rhomboids and levator scapulae are supplied by the:

Answer: B) Dorsal scapular nerve

The dorsal scapular nerve (C5) supplies the rhomboids and levator scapulae.

81. Deep (intrinsic) back muscles such as erector spinae are innervated by the:

Answer: B) Dorsal primary rami

Intrinsic back muscles are supplied by the dorsal (posterior) primary rami of spinal nerves.

82. Upper cervical transverse-process fractures (C1–C4) could weaken the:

Answer: E) Levator scapulae

Levator scapulae originates from the transverse processes of C1–C4.

83. Spondylolysis is most commonly a fracture/defect of the:

Answer: A) Pars interarticularis

Spondylolysis is a stress fracture/defect of the pars interarticularis of the vertebral arch.

84. The ROOF of the suboccipital triangle is formed by the:

Answer: C) Semispinalis capitis

Semispinalis capitis forms the roof; the three suboccipital muscles form the boundaries.

85. A hyperextension (whiplash) injury most often tears the:

Answer: D) Anterior longitudinal ligament

Hyperextension stretches the anterior column, tearing the anterior longitudinal ligament.

86. The contents of the suboccipital triangle are the:

Answer: C) Vertebral artery and suboccipital nerve

The triangle contains the vertebral artery and the suboccipital nerve (dorsal ramus of C1).

87. A stab wound to the spinal accessory nerve in the posterior triangle paralyses the:

Answer: E) Trapezius

CN XI is superficial in the posterior triangle and supplies trapezius (and sternocleidomastoid).

88. The suboccipital nerve (dorsal ramus of C1) spares which muscle?

Answer: A) Splenius capitis

The suboccipital nerve supplies the four suboccipital muscles; splenius capitis is a more superficial back muscle of other dorsal rami.

89. The lumbar (Petit) triangle is bounded by latissimus dorsi, iliac crest and the:

Answer: E) External abdominal oblique

Borders: latissimus dorsi (medial), external oblique (lateral), iliac crest (inferior).

90. Erector spinae columns from lateral to medial are:

Answer: E) Iliocostalis, longissimus, spinalis

Lateral→medial: Iliocostalis, Longissimus, Spinalis.

91. The transversospinalis group from superficial to deep is:

Answer: E) Semispinalis, multifidus, rotatores

Superficial→deep: semispinalis, multifidus, rotatores.

92. Rotation of the head ('NO' movement) occurs mainly at the:

Answer: C) Median atlanto-axial joint

The median atlanto-axial (pivot) joint rotates the head; the dens of C2 is held by the transverse ligament.

93. The atlanto-occipital joint chiefly permits:

Answer: E) Flexion/extension nodding

The atlanto-occipital joint gives the flexion/extension 'YES' nod.

94. Approximately 95% of lumbar disc herniations occur at:

Answer: C) Levels L4–L5 or L5–S1

Most lumbar disc protrusions are at L4–L5 or L5–S1, causing sciatica.

95. Rupture of the transverse ligament of the atlas leads to:

Answer: C) Atlanto-axial subluxation

If the transverse ligament ruptures, the dens is freed and the median atlanto-axial joint subluxes, threatening the cord.

96. The nucleus pulposus of the intervertebral disc is:

Answer: A) The semifluid central core

The nucleus pulposus is the semifluid core; the anulus fibrosus is the surrounding fibrocartilage ring.

97. The ligament that limits flexion and is elastic, helping straighten the column, is the:

Answer: D) Ligamentum flavum

Ligamentum flavum (elastic) connects laminae, limits abrupt flexion and assists straightening.

98. Spondylolisthesis is best described as:

Answer: A) Forward vertebral slippage

Spondylolisthesis is anterior slippage of a vertebral body on the one below (often after bilateral pars defects).

Section 4 · Scapular, Deltoid, Pectoral & Axilla (Lec 10–11)

99. A fracture of the bony 'point' of the shoulder involves the:

Answer: E) Acromion

The acromion forms the bony summit ('point') of the shoulder.

100. Posterior-cord injury of the brachial plexus chiefly affects the:

Answer: D) Radial nerve

The radial nerve is the main continuation of the posterior cord; cord injury has wide posterior-compartment effects.

101. Which muscle initiates abduction of the arm (0–15°)?

Answer: E) Supraspinatus

Supraspinatus initiates the first ~15°; deltoid then takes over.

102. Upper-root (C5–C6, Erb) brachial-plexus injury spares the:

Answer: A) Flexor carpi ulnaris

Flexor carpi ulnaris is ulnar (C8–T1); the C5–C6 muscles (deltoid, biceps, brachialis, supraspinatus) are affected.

103. Excision of a mole just above the clavicle risks cutting the:

Answer: A) Supraclavicular nerves

The supraclavicular nerves (C3–C4) are superficial over the clavicle/upper shoulder.

104. Which axillary node group mainly drains the breast?

Answer: D) The pectoral (anterior) group

The pectoral (anterior) group along pectoralis minor receives most breast lymph.

105. After modified mastectomy, weakness of the LOWER pectoralis major indicates injury to the:

Answer: B) Medial pectoral nerve

The medial pectoral nerve supplies the sternocostal (lower) head of pectoralis major and pectoralis minor.

106. Mid-shaft clavicle fracture with loss of elbow flexion suggests injury to the:

Answer: C) The lateral cord (musculocut.)

Loss of elbow flexion implicates the musculocutaneous nerve, a lateral-cord branch passing deep to the clavicle.

107. A winged scapula (medial border lifts off) indicates injury to the:

Answer: B) Long thoracic nerve

Serratus anterior (long thoracic nerve) stabilises the scapula; its palsy causes medial winging.

108. Anterior–inferior shoulder dislocation most endangers the:

Answer: A) Axillary nerve

The axillary nerve (with the posterior circumflex humeral artery) crosses the surgical neck/quadrangular space and is at risk.

109. Dorsal scapular-nerve injury weakens which muscle?

Answer: E) Rhomboid major

The dorsal scapular nerve (C5) supplies the rhomboids (and levator scapulae).

110. The cephalic vein runs in the deltopectoral groove and drains into the:

Answer: A) Axillary vein

The cephalic vein pierces the clavipectoral fascia to join the axillary vein.

Section 5 · Arm, Forearm & Cubital Fossa (Lec 12–13)

111. Which nerve lies posterior to the medial epicondyle of the humerus?

Answer: E) Ulnar nerve

The ulnar nerve grooves behind the medial epicondyle (the 'funny bone'), where it is superficial.

112. The lateral cutaneous nerve of the forearm is the continuation of the:

Answer: A) The musculocutaneous nerve

After supplying the anterior arm muscles, the musculocutaneous nerve continues as the (sensory) lateral cutaneous nerve of the forearm.

113. An absent biceps reflex indicates a lesion at which spinal level?

Answer: C) C5

The biceps reflex tests C5–C6 (musculocutaneous), with C5 the primary root.

114. Which muscle medially rotates the arm?

Answer: C) Latissimus dorsi

Latissimus dorsi adducts, extends and medially rotates the humerus; infraspinatus/teres minor laterally rotate.

115. Which cutaneous nerve of the upper limb is NOT from the posterior cord?

Answer: C) Lateral cutaneous of forearm

The lateral cutaneous nerve of the forearm comes from the musculocutaneous (lateral cord); the others are axillary/radial (posterior cord).

116. Fracture of which humeral part may injure the ulnar nerve?

Answer: A) Medial epicondyle

A medial-epicondyle fracture can damage the ulnar nerve in its groove.

117. A high median-nerve injury at the elbow chiefly causes:

Answer: E) Weak forearm flexion/pronation

High median injury paralyzes most forearm flexors and the pronators, causing weakness of flexion/pronation.

118. Among brachial-plexus branches, which supplies an EXTENSOR of the arm?

Answer: A) Thoracodorsal

Thoracodorsal supplies latissimus dorsi, a powerful extensor of the arm at the shoulder.

119. Olecranon fracture causes weakness of elbow extension because it detaches the:

Answer: B) Triceps insertion

The olecranon is the triceps insertion; fracture there weakens extension.

120. In the cubital fossa, the order from superficial to deep is:

Answer: A) Skin, vein, aponeurosis, artery

Superficial→deep: skin/superficial fascia (median cubital vein) → bicipital aponeurosis → brachial artery → median nerve.

121. A lateral-epicondyle (common extensor origin) fracture SPARES which muscle?

Answer: B) Extensor indicis

Extensor indicis arises distally from the ulna/interosseous membrane, not the common extensor origin.

122. For surgery at the SURGICAL NECK of the humerus, the preferred block is:

Answer: E) Interscalene block

An interscalene block (roots/trunks) covers the shoulder/proximal humerus, including the axillary nerve.

123. For mid-shaft humerus and distal surgery, a suitable block is the:

Answer: C) Axillary block

An axillary block anaesthetises the terminal branches for forearm/hand and mid-humerus-distal surgery.

124. Klumpke palsy (lower trunk C8–T1) classically produces:

Answer: A) Ulnar claw hand

Lower-trunk injury affects intrinsic hand muscles, giving an ulnar claw hand ($\pm$ Horner's).

125. Pain on resisted elbow flexion in SUPINATION points to injury of the:

Answer: E) Biceps brachii

Biceps is the strong flexor/supinator; resisted flexion in supination loads it maximally.

126. Inability to supinate the forearm is most often accompanied by:

Answer: C) Weak elbow extension

Both the supinator and (proximally) triceps are radial-nerve muscles, so radial injury links lost supination with weak elbow extension.

127. Which forearm muscle–nerve pairing is CORRECT?

Answer: D) EI — posterior interosseous

Extensor indicis is supplied by the posterior interosseous nerve (deep radial). FPL is AIN/median; pronator teres median; FCU ulnar; brachioradialis radial.

128. A spiral (mid-shaft) humeral fracture classically causes:

Answer: D) Wrist drop

The radial nerve in the spiral groove is injured, causing wrist drop (loss of wrist/finger extension).

129. 'Golfer's elbow' (medial epicondylitis) involves the:

Answer: C) Common flexor origin

Medial epicondylitis affects the common flexor origin from repetitive wrist flexion; tennis elbow affects the common extensor (ECRB).

130. 'Saturday-night palsy' produces:

Answer: E) Wrist drop

Radial-nerve compression against the humerus causes wrist drop.

131. A Colles' fracture is a distal radius fracture with:

Answer: D) Posterior (dorsal) displacement

Colles' fracture shows dorsal (posterior) displacement of the distal fragment, classically from a fall on an outstretched hand.

132. The brachial artery divides into radial and ulnar arteries at the:

Answer: B) Apex of the cubital fossa

The brachial artery bifurcates at the apex of the cubital fossa (neck of radius).

Section 6 · Hand & Joints of Upper Limb (Lec 14–15)

133. The radio-ulnar interosseous membrane is a:

Answer: A) Syndesmosis

It is a fibrous syndesmosis uniting radius and ulna.

134. Which ligament reinforces the sternoclavicular joint and limits girdle elevation?

Answer: D) Costoclavicular ligament

The costoclavicular ligament anchors the medial clavicle to the first rib, limiting elevation and stabilising the SC joint.

135. A wrist injury gives weak thumb flexion, abduction and inability to oppose, with NO sensory loss. The injured structure is the:

Answer: D) Recurrent branch of median nerve

The recurrent (motor) branch of the median nerve supplies the thenar muscles; isolated thenar weakness without sensory loss localises here.

136. A claw-hand deformity signals injury to the:

Answer: C) Ulnar nerve

Ulnar palsy paralyse interossei and medial two lumbricals $\rightarrow$ claw hand of the 4th/5th digits.

137. Anterior shoulder dislocation does NOT injure which structure (it stabilises the AC, not GH, joint)?

Answer: A) Coracoclavicular ligament

The coracoclavicular ligament stabilises the acromioclavicular joint, not the glenohumeral joint.

138. The carpal tunnel transmits the median nerve and:

Answer: B) Flexor tendons

The carpal tunnel transmits the median nerve and the long flexor tendons under the flexor retinaculum.

139. On the ulnar side, the flexor retinaculum attaches to the:

Answer: A) Pisiform and hook of hamate

Ulnar attachment: pisiform and hook of hamate; radial attachment: scaphoid tubercle and trapezium ridge.

140. Most intrinsic hand muscles are supplied by the deep branch of the ulnar nerve EXCEPT the thenar muscles and the:

Answer: C) Lateral two lumbricals

The lateral two lumbricals (and the three thenar muscles) are median; the rest are deep ulnar.

141. The prime mover of thumb opposition is the:

Answer: E) Opponens pollicis

Opponens pollicis is the prime mover of opposition at the thumb saddle (carpometacarpal) joint.

142. The dorsal interossei of the hand act to:

Answer: E) Abduct the fingers

Dorsal interossei ABduct fingers (DAB); palmar interossei ADduct (PAD).

143. The lumbricals flex the MCP joints and:

Answer: B) Extend the IP joints

Lumbricals (from FDP tendons to the extensor expansions) flex MCP and extend IP joints.

144. The elbow (humero-ulnar) joint is a:

Answer: E) Hinge joint

The elbow is a uniaxial hinge joint (flexion/extension).

145. The wrist (radiocarpal) joint is a:

Answer: D) A condyloid (ellipsoid) joint

The radiocarpal joint is condyloid, allowing flexion/extension and abduction/adduction.

146. The proximal radio-ulnar joint is a:

Answer: D) Pivot joint

It is a pivot joint (radial head rotating within the annular ligament) for pronation/supination.

147. The 1st carpometacarpal (thumb) joint is a:

Answer: A) Saddle joint

The thumb CMC joint (trapezium–metacarpal I) is a saddle joint enabling opposition.

148. The glenoid cavity is deepened by the:

Answer: E) Glenoid labrum

The fibrocartilaginous glenoid labrum deepens the glenoid for stability.

149. The four rotator cuff muscles are supraspinatus, infraspinatus, teres minor and:

Answer: C) Subscapularis

SITS = Supraspinatus, Infraspinatus, Teres minor, Subscapularis. Teres major is NOT part of the cuff.

150. 'Nursemaid's elbow' (radial-head subluxation) involves the:

Answer: D) Annular ligament

Axial traction slips the radial head from under the loose annular ligament in young children.

151. The chief flexor and chief extensor of the elbow are, respectively:

Answer: C) Brachialis; triceps brachii

Per the slide, brachialis is the chief flexor and triceps brachii the chief extensor of the elbow.

152. The anatomical snuffbox floor is formed by the scaphoid and:

Answer: C) Trapezium

The snuffbox floor is the scaphoid and trapezium; its content is the radial artery (2nd part).

Section 7 · Development of Bone, Skin & Muscle (Lec 16–17)

153. Premature fusion of the SAGITTAL suture produces a long, narrow skull termed:
Answer: B) Scaphocephaly
Sagittal synostosis limits width, giving a long boat-shaped skull (scaphocephaly).
154. Premature bilateral fusion of the CORONAL sutures gives a short, broad skull termed:
Answer: E) Brachycephaly
Bilateral coronal synostosis limits AP growth → brachycephaly.
155. Vertebrae and ribs develop from the:
Answer: D) Sclerotome
The sclerotome forms the axial skeleton (vertebrae, ribs).
156. Skeletal muscle of the limbs develops from the:
Answer: E) Myotome
The myotome of the somite forms skeletal muscle.
157. The flat bones of the skull vault form by:
Answer: D) Intramembranous ossification
Vault bones (frontal, parietal) form by intramembranous ossification within mesenchyme.
158. The skull base (sphenoid, ethmoid, occipital) forms by:
Answer: C) Endochondral ossification
The chondrocranium (skull base) ossifies endochondrally from a cartilage model.
159. Congenital absence of the sternocostal head of pectoralis major with ipsilateral hand defects is:
Answer: C) Poland syndrome
Poland syndrome = absent/hypoplastic pectoralis major (sternocostal head) with ipsilateral limb anomalies.
160. Partial/complete congenital absence of abdominal-wall musculature characterises:
Answer: B) Prune-belly syndrome
Prune-belly (Eagle-Barrett) syndrome features deficient abdominal-wall muscle.
161. Fusion/webbing of adjacent digits is termed:
Answer: E) Syndactyly
Syndactyly is failure of separation (webbing) of digits.
162. Bilateral clubfoot (talipes equinovarus) centres on a deformity involving the:
Answer: C) Talus
Clubfoot involves a talar deformity, with the foot plantarflexed and inverted (equinovarus).

Section 8 · Gluteal Region, Posterior Thigh & Popliteal Fossa (Lec 18–19)

163. The safe quadrant for a deep IM gluteal injection (avoiding the sciatic nerve) is the:
Answer: C) Superolateral
The superolateral (upper-outer) quadrant avoids the inferomedially-running sciatic nerve.
164. Gluteus maximus is innervated by the:
Answer: C) Inferior gluteal nerve
Gluteus maximus (extensor, lateral rotator of the thigh) is supplied by the inferior gluteal nerve.
165. Gluteus medius and minimus are supplied by the:
Answer: B) Superior gluteal nerve
The superior gluteal nerve supplies gluteus medius, minimus and tensor fasciae latae.
166. Paralysis of gluteus medius/minimus produces a positive:
Answer: E) Trendelenburg sign
Loss of hip abductor stabilisation causes the unsupported pelvis to drop — positive Trendelenburg sign.
167. Gluteus maximus chiefly produces which action at the hip?
Answer: B) Extension, lateral rotation
Gluteus maximus extends and laterally rotates the thigh (important for rising and climbing).
168. Gluteus medius and minimus together produce abduction and:
Answer: B) Medial thigh rotation
They abduct and medially rotate the thigh and stabilise the pelvis in stance.

169. A posterior hip dislocation most often injures the:

Answer: C) Sciatic nerve

The sciatic nerve lies posterior to the hip and is endangered by posterior dislocation, often causing foot drop.

170. Avascular necrosis of the femoral head most often follows damage to the:

Answer: A) Medial circumflex femoral artery

The medial circumflex femoral artery supplies most of the femoral head via retinacular branches.

171. In the popliteal fossa, the MOST superficial central structure is the:

Answer: E) Tibial nerve

Superficial→deep: tibial nerve → popliteal vein → popliteal artery (deepest).

172. Which muscle is NOT a 'true hamstring' (supplied by the common fibular division)?

Answer: D) Short head of biceps femoris

The short head of biceps femoris is supplied by the common fibular division, so it is not a true (tibial-division) hamstring.

173. Which structure exits the greater sciatic foramen ABOVE piriformis?

Answer: E) Superior gluteal nerve

The superior gluteal nerve/vessels exit above piriformis; most others exit below.

174. The gluteal ligaments that convert the sciatic notches into foramina are the sacrospinous and:

Answer: D) The sacrotuberous ligament

The sacrotuberous and sacrospinous ligaments form the greater and lesser sciatic foramina.

Section 9 · Anterior/Medial Thigh, Leg, Foot & Lower-limb Joints (Lec 20–23)

175. The anterior compartment of the thigh (quadriceps) is innervated by the:

Answer: A) Femoral nerve

The femoral nerve supplies the quadriceps (knee extension) and gives the saphenous nerve.

176. The medial (adductor) compartment of the thigh is innervated by the:

Answer: E) Obturator nerve

The obturator nerve supplies the adductors, gracilis and obturator externus, plus medial-thigh skin.

177. The anterior compartment of the leg is supplied by the:

Answer: D) Deep fibular nerve

The deep fibular nerve and anterior tibial artery supply the anterior leg (dorsiflexors).

178. The lateral compartment of the leg is supplied by the:

Answer: A) Superficial fibular nerve

The superficial fibular nerve supplies fibularis longus and brevis (evertors).

179. The posterior compartment of the leg is supplied by the:

Answer: A) Tibial nerve

The tibial nerve and posterior tibial artery supply the calf (plantarflexors).

180. The main action of the anterior leg compartment is:

Answer: A) Dorsiflexion

The anterior compartment (tibialis anterior, EDL, EHL, fibularis tertius) dorsiflexes the foot.

181. The lateral leg compartment produces:

Answer: A) Eversion

Fibularis longus/brevis evert the foot (with some plantarflexion).

182. The triceps surae (gastrocnemius + soleus) act to:

Answer: C) Plantarflex the ankle

The triceps surae plantarflex the ankle via the calcaneal tendon for push-off.

183. The muscle that 'unlocks' the fully extended knee is the:

Answer: D) Popliteus

Popliteus laterally rotates the femur on the fixed tibia to unlock the knee and begin flexion.

184. A fracture at the neck of the fibula most commonly injures the:

Answer: C) Common fibular nerve

The common fibular nerve winds around the fibular neck and is the most commonly injured lower-limb nerve (foot drop).

185. Foot drop (loss of dorsiflexion and eversion) follows injury to the:

Answer: A) Common fibular nerve

Common fibular injury knocks out dorsiflexors (deep branch) and evertors (superficial branch) → foot drop.

186. Tarsal-tunnel syndrome compresses the:

Answer: D) Tibial nerve

The tibial nerve passes the tarsal tunnel behind the medial malleolus; compression gives sole numbness and weak intrinsic.

187. Tibialis posterior chiefly produces:

Answer: C) Plantarflexion and inversion

Tibialis posterior plantarflexes and inverts the foot and supports the medial arch.

188. Collapse of the medial longitudinal arch (flatfoot) classically follows dysfunction of the:

Answer: D) Tibialis posterior tendon

Tibialis posterior is the dynamic supporter of the medial arch; its dysfunction causes acquired flatfoot.

189. The hip joint is which type of synovial joint?

Answer: B) Ball-and-socket

The hip is a multiaxial ball-and-socket joint.

190. The strongest hip ligament, limiting hyperextension, is the:

Answer: A) Iliofemoral ligament

The Y-shaped iliofemoral ligament (AIIIS to intertrochanteric line) is the strongest and limits hyperextension.

191. The knee is best described as a:

Answer: D) Modified hinge joint

The knee is a modified hinge (allowing some rotation), with patellofemoral and tibiofemoral articulations.

192. The anterior cruciate ligament arises from which area of the tibia?

Answer: B) Anterior intercondylar area

The ACL arises from the anterior intercondylar area and prevents anterior tibial displacement.

193. The posterior cruciate ligament prevents:

Answer: A) Posterior tibial displacement

The PCL (from the posterior intercondylar area) prevents posterior displacement of the tibia.

194. Which meniscus is C-shaped and firmly attached to the tibial collateral ligament?

Answer: E) Medial meniscus

The medial meniscus is C-shaped and adherent to the tibial collateral ligament, so it tears more often.

195. The acetabulum is formed by ilium, ischium and:

Answer: E) Pubis

Ilium, ischium and pubis meet at the triradiate cartilage to form the acetabulum.

196. The great saphenous vein ultimately drains into the:

Answer: B) Femoral vein

The great saphenous vein traverses the saphenous opening to drain into the femoral vein.

197. The small saphenous vein drains into the:

Answer: D) Popliteal vein

The small saphenous vein pierces the deep fascia to empty into the popliteal vein.

198. At heel-strike, the ankle is held in:

Answer: E) Dorsiflexion

At heel-strike the hip is flexed, knee extended and ankle dorsiflexed as the heel contacts.

199. The popliteal artery divides into the anterior tibial and:

Answer: C) Posterior tibial artery

The popliteal artery divides into anterior tibial and posterior tibial arteries; the latter gives the fibular artery.

200. Collapse-resistant 'close-packed' stability of the ankle occurs in:

Answer: D) Dorsiflexion

The talus is wider anteriorly; in dorsiflexion it wedges into the mortise — most stable position.

201. The hindfoot consists of the talus and:

Answer: C) Calcaneus

Hindfoot = talus + calcaneus; midfoot = navicular, cuboid, cuneiforms; forefoot = metatarsals + phalanges.

202. The plantar aponeurosis arises posteriorly from the:

Answer: B) Calcaneus

The plantar aponeurosis (thick central plantar fascia) arises from the calcaneus and divides into five digital bands.

Examiner patterns to expect: repeated core concepts across versions; 'which is NOT/false/spares' phrasing; nerve–muscle–action pairings; joint-type classification; histology cell/stain ID; collagen-type matching; landmark triangles & their borders; compartment→nerve→artery→action grouping; and development/syndrome terms. Reason from the deficit to the structure rather than memorising wording.